Teacher Wu Kuan | Midterm Must-Miss #4 — Parallel Lines & Similar Triangles

English lecture notes · YouTube Short 6C3Ufsn8lKI

1. The trap question

DE ∥ BC, AD = 4, DB = 2, BC = 9. Find DE.

2. The method

• DE ∥ BC → △ADE ~ △ABC.
• Match small to whole: DE : BC = AD : AB (not AD : DB).
• AB = 6 → DE = 9 × 4/6 = 6. Sanity check: DE must be shorter than BC.

3. Parent check method

Ask your child to trace the small triangle and the big triangle in two colors before writing the ratio.

Practice

1. DE ∥ BC, AD = 3, DB = 6, BC = 12. Find DE.
2. DE ∥ BC, AD = 2, AB = 8, BC = 10. Find DE.
3. DE ∥ BC, AD = DB = 5, DE = 7. Find BC.
4. DE ∥ BC, AE = 6, EC = 3, BC = 12. Find DE.
5. DE ∥ BC, AD : DB = 2 : 3, DE = 4. Find BC.

Answers

1. 4
2. 2.5
3. 14
4. 8
5. 10

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